COOLING TIME IN INJECTION MOULDING

BALA
THANZELLA
DHANYA
VIGNESH
RAMANA

OUTPUT TO BE FOUND

Optimizing cycle time

Reducing defects like warping or shrinkage

Improving product consistency

Enhancing overall manufacturing efficiency

DATA FEATURES

- Mold material
- Product thickness (mm)
- Coolant temperature (°C)
- Injection pressure (bar)
- Ambient temperature (°C)
- Cycle time (s)
- Cavity size (cm^3)

REGRESSION ALGORITHM

REGRESSION ALGORITHMS USED:

LINEAR REGRESSION – BASE MODEL FOR COMPARISON

RIDGE REGRESSION – L2 REGULARIZATION TO REDUCE OVERFITTING

LASSO REGRESSION – L1 REGULARIZATION, ALSO PERFORMS FEATURE SELECTION

ELASTICNET REGRESSION – COMBINES L1 AND L2 REGULARIZATION

BEST ALGORITHM

BEST MODEL: LASSO REGRESSION
WHY LASSO?

EXCELLENT PREDICTION ACCURACY ($R^2 \approx 1$)

PERFORMS AUTOMATIC FEATURE SELECTION

REDUCES OVERFITTING WITH L1 REGULARIZATION

SIMPLIFIES THE MODEL FOR BETTER INTERPRETATION

Benefits for predicting

1. Increased Productivity

Shorter Cycle Times: Faster cooling means shorter cycles, which allows more parts to be produced in the same amount of time.

Higher Throughput: More efficient use of machines and labor, lowering cost per part.

2. Improved Dimensional Stability

Controlled Shrinkage: Optimized cooling reduces warping and internal stresses, leading to better dimensional accuracy.

Consistent Part Quality: Even cooling avoids defects like sink marks and voids.

THANK YOU